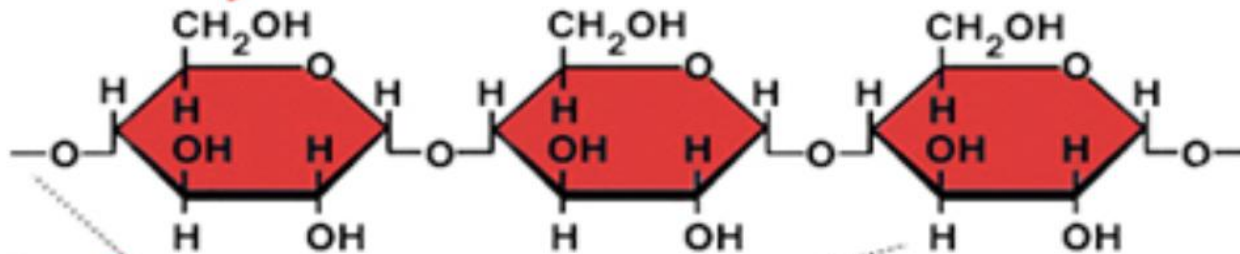
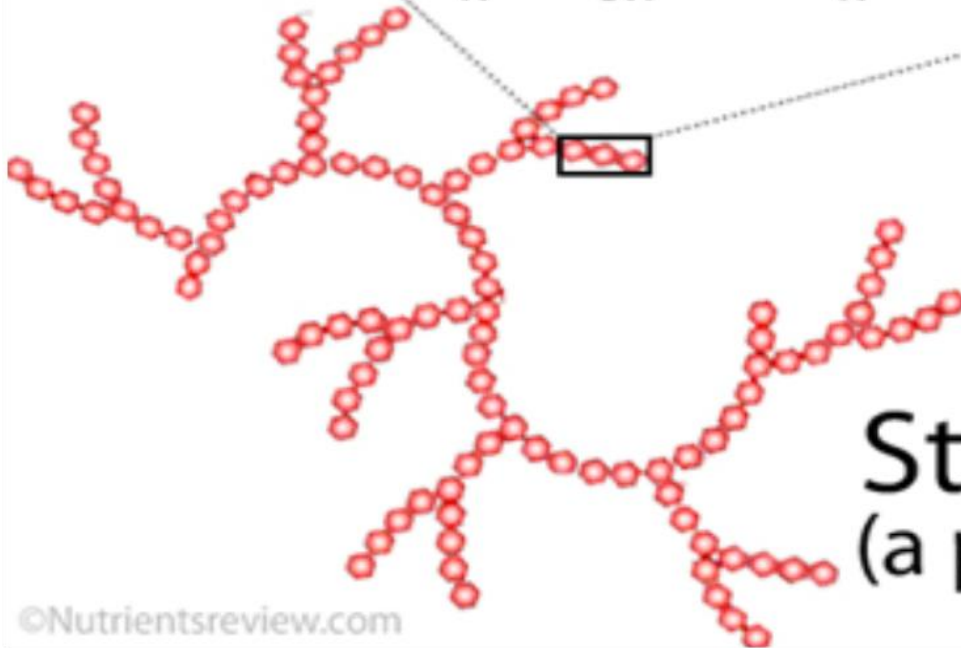


Polysaccharides



Glucose
molecules



Starch
(a polysaccharide)

POLYSACCHARIDES

- ❖ Many monosaccharides may combine by **condensation** reactions to form polysaccharides.
- ❖ A number of monosaccharides which combine may be variable and the chain can be **branched** or **unbranched**.

Properties of polysaccharides include:

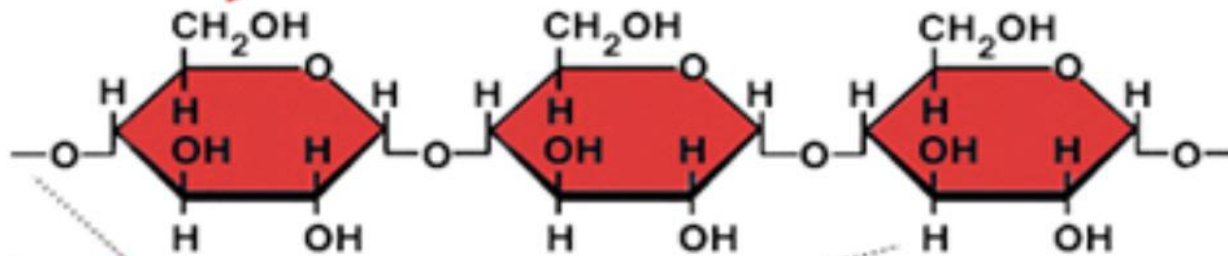
Are not sweet

- ❖ *Do not dissolve in water*
- ❖ *Cannot be crystallized*
- ❖ *They have a high molecular mass.*
- ❖ *They are non-reducing sugars*
- ❖ *The chains may be folded to make them compact which are ideal for storage. Such a large size of the molecules makes them insoluble in water and suitable for storage as they exert no osmotic influence and do not easily diffuse out of the cell.*

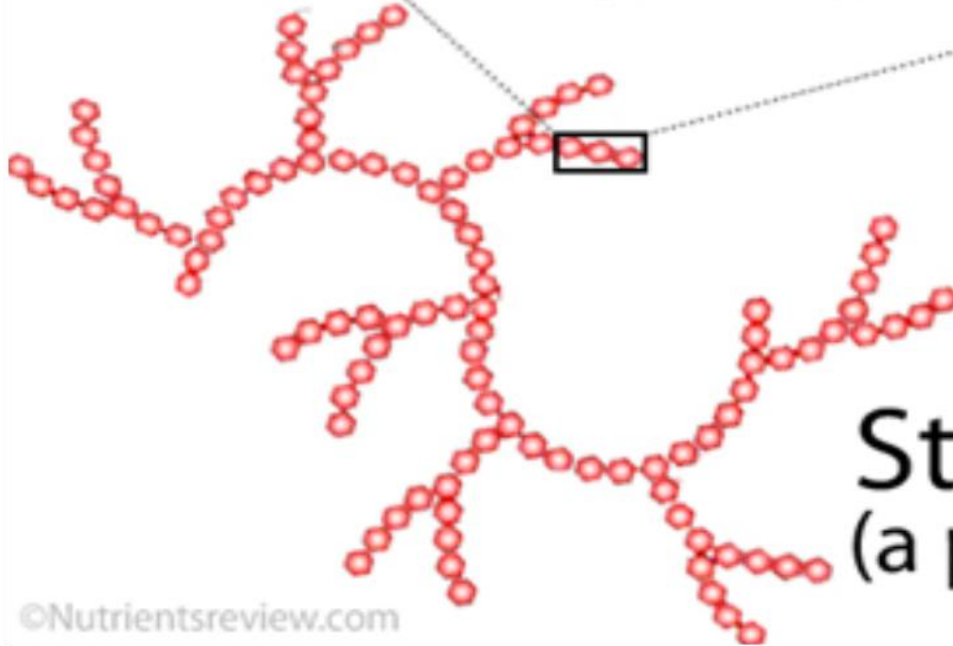
Properties of polysaccharides cont..

- ❖ **Starch** is the main storage material in green **plants** while **glycogen** is for **animals**.
- ❖ Upon **hydrolysis**, polysaccharides are broken down into their constituent *monosaccharides*.
- ❖ Not all polysaccharides are used for storage e.g. **cellulose** is a structural polysaccharide giving **support** and strength to the cell walls.

Polysaccharides



Glucose
molecules



Starch
(a polysaccharide)

STARCH

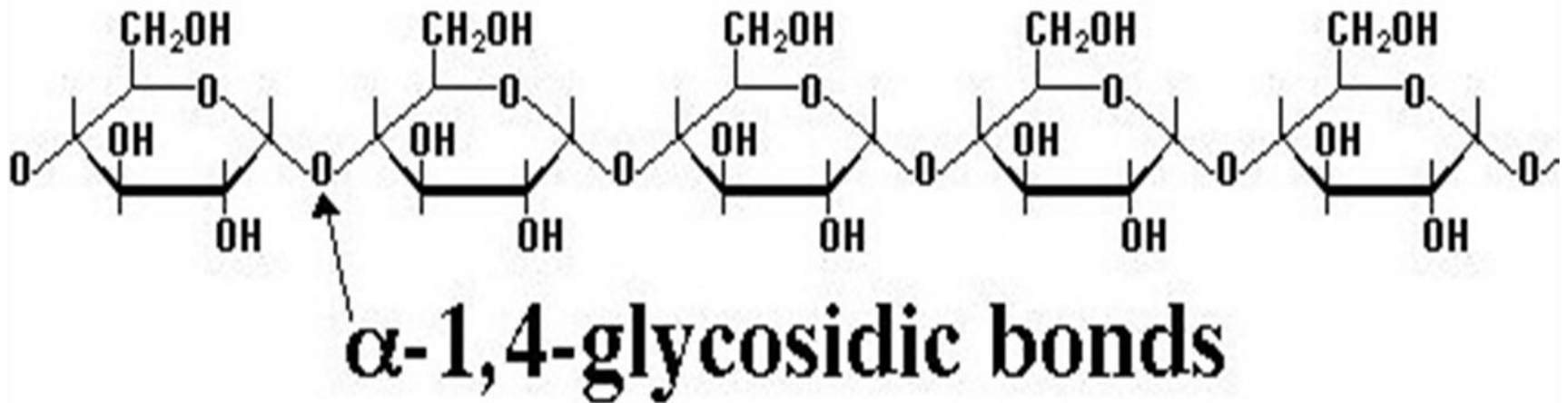
- ❖ It is found in plant parts in form of granules. It is a reserve food formed from any excess glucose during photosynthesis.
- ❖ It is common in seeds e.g. maize where it is the main food supply during germination.

Structure of starch

- ❖ It is a **polymer** of **α -glucose** molecules which are held by glycosidic bonds forming chains of α -glucose units which get folded or coiled into **a helix**.
- ❖ Starch has two components i.e. **amylose** and **amylopectin**, that is, **20%** amylose, **79%** amylopectin and **1%** other substances e.g. phosphates and fatty acids.
- ❖ Amylose stains deep blue with iodine while amylopectin stains red to purple with iodine.
- ❖ **Amylose** is structurally **unbranched** while amylopectin is branched.

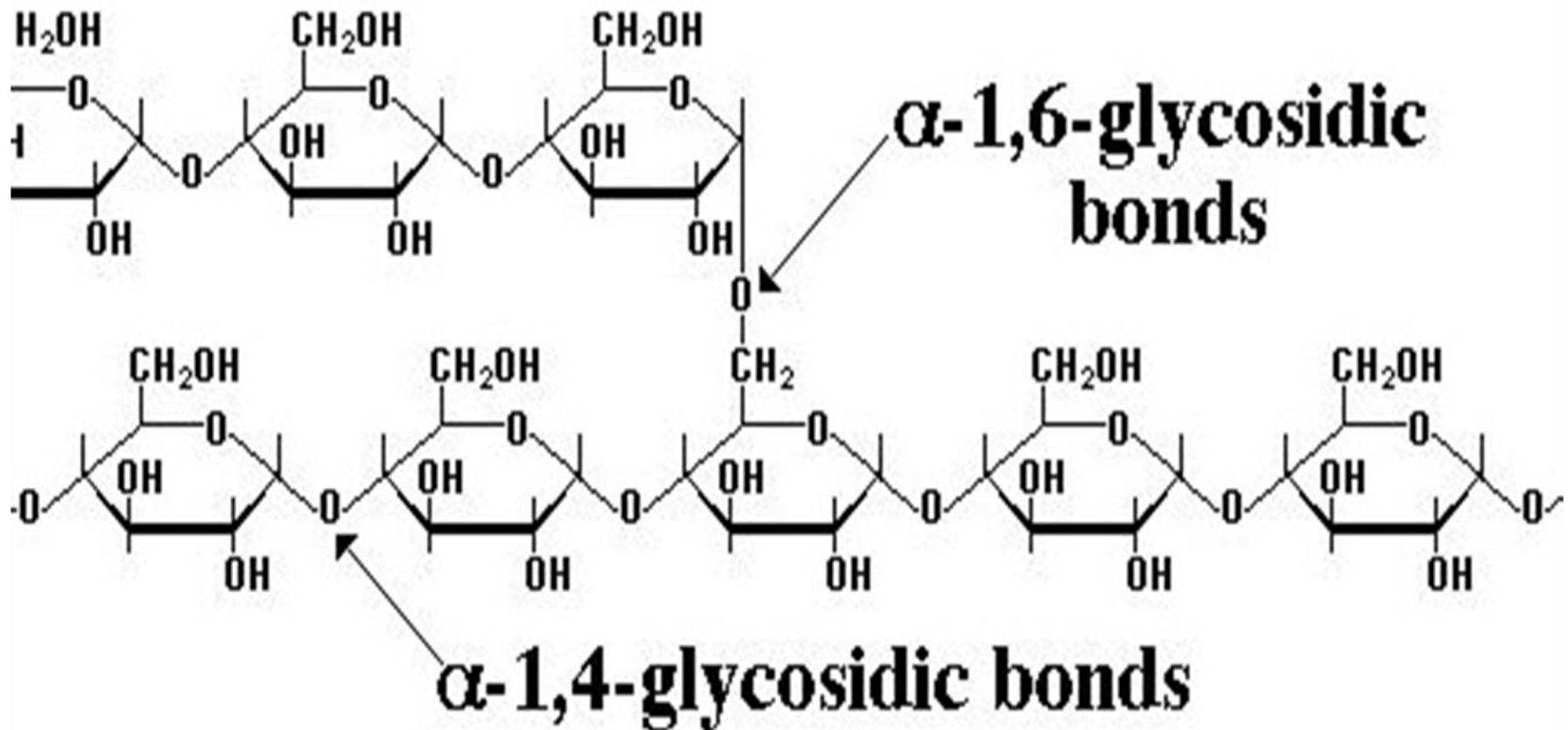
STRUCTURE OF AMYLOSE

Amylose

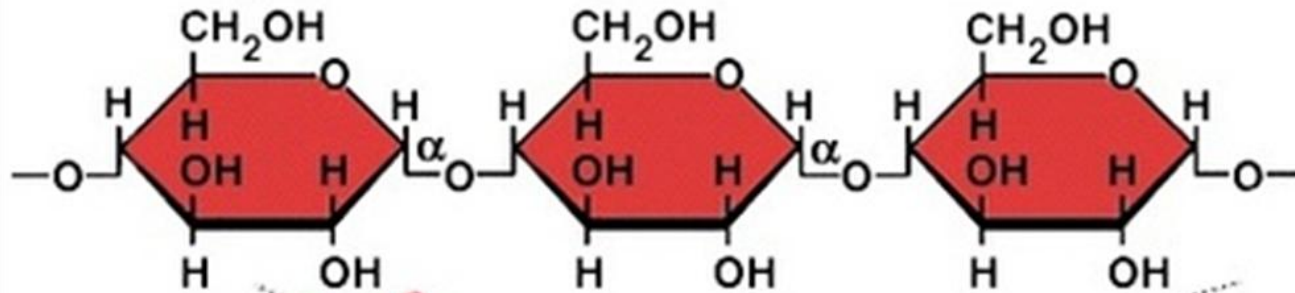


STRUCTURE OF AMYLOPECTIN

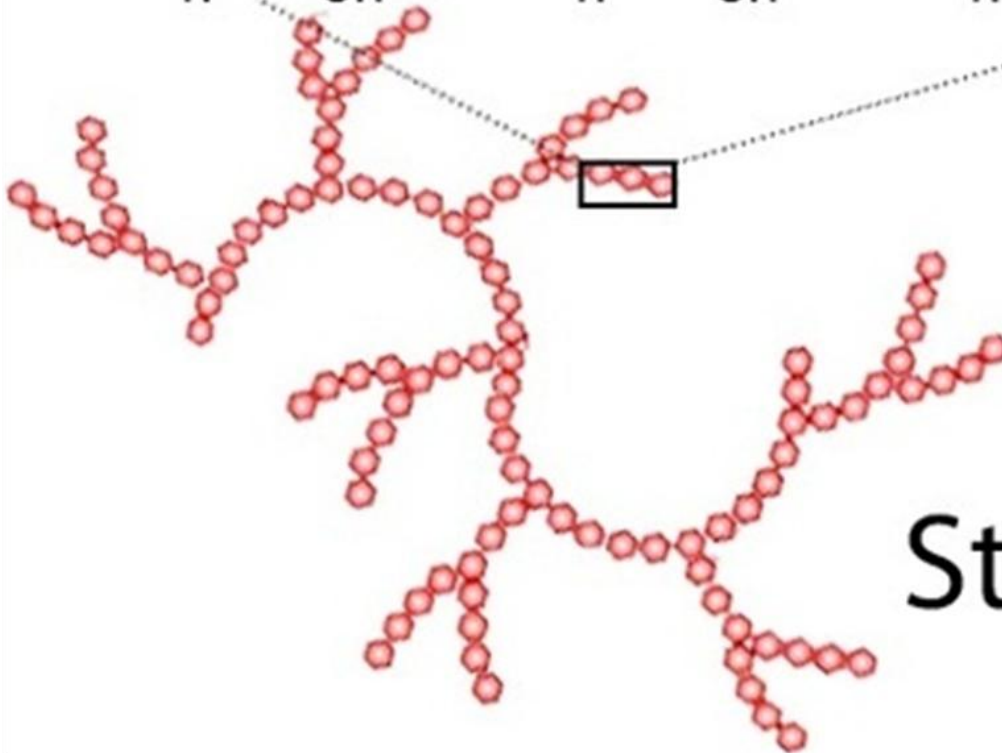
Amylopectin



Starch



Glucose
molecules



Starch

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QUESTION:

***HOW DOES THE
STRUCTURE OF STARCH
RELATE TO ITS ROLES?***

Solution

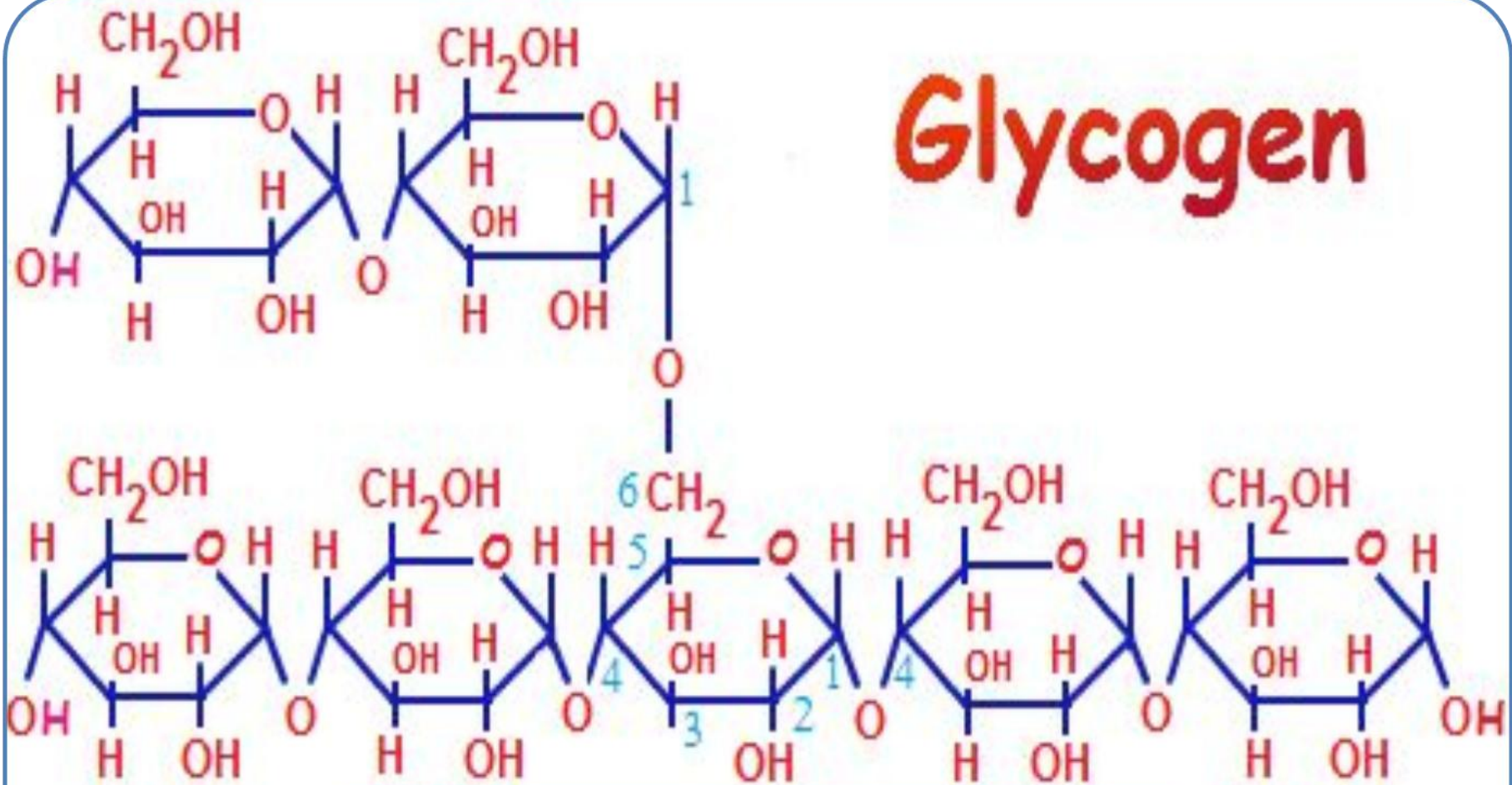
- ❖ It is **a polymer** of α -glucose molecules hence a large molecule making it relatively **insoluble** in water hence an ideal storage molecule.
- ❖ The α -glucose molecules are held by **glycosidic bonds** which can be broken down to free glucose molecules from the stored starch for ATP synthesis during respiration.
- ❖ The starch molecule is coiled into **a helix** with a hydroxide group projecting interiorly making it insoluble in water hence exerts no osmotic effects in cells and is ideal for storage.
- ❖ The chains of α -glucose units can **be folded** to form starch grains for easy storage.

GLYCOGEN

It is a major polysaccharide storage material in animals. It stored mainly in the liver and muscles. It is also made up of **α-glucose** molecules and exists as granules.

- ❖ However its chains are shorter (10-20 glucose units) and is more branched.
- ❖ Glycogen is more soluble than starch

Glycogen



Structure of glycogen

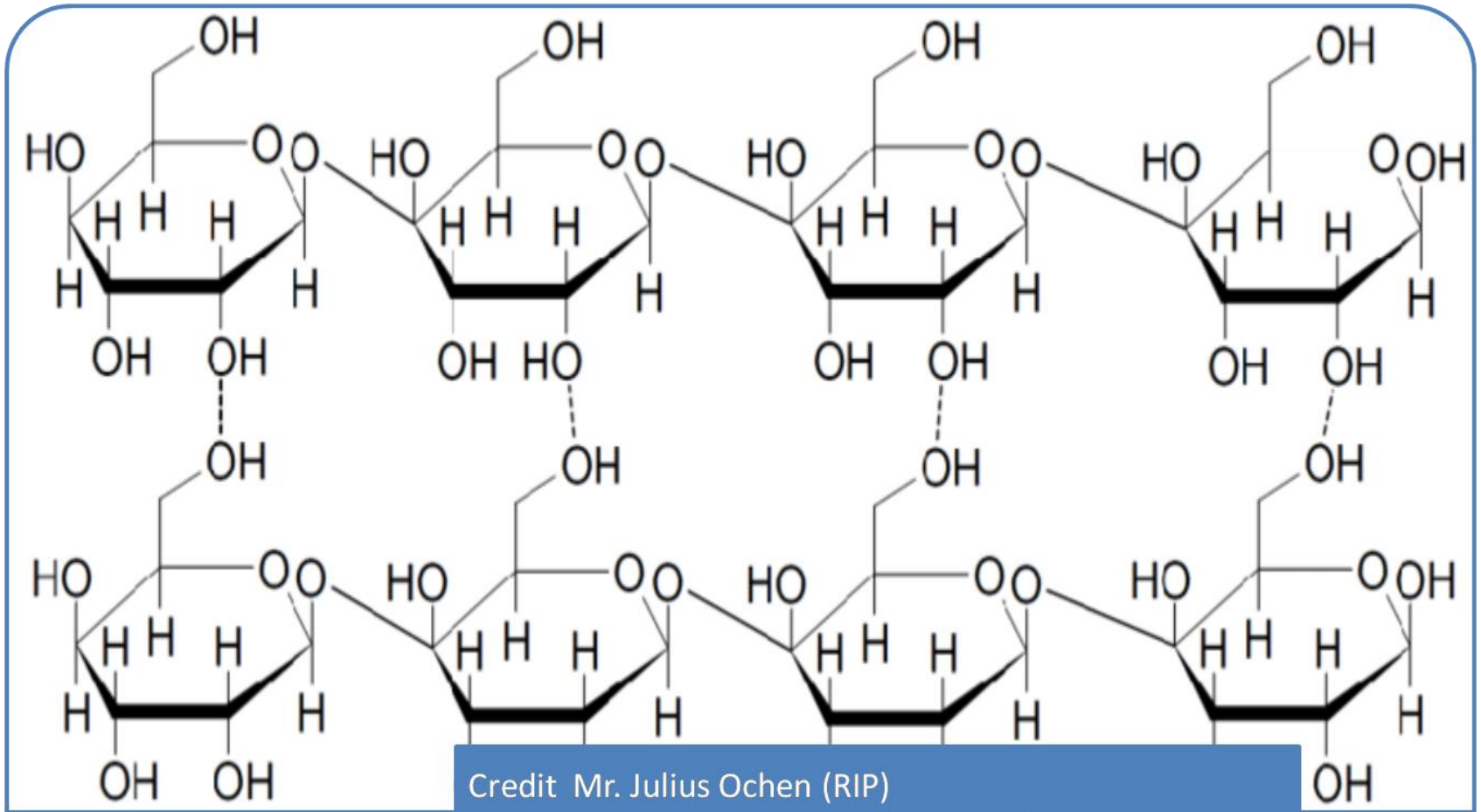
CELLULOSE

- ❖ It comprises up to **50%** of a plant cell wall and in cotton it makes up to 90%.
- ❖ It is a polymer of about 10000 β -glucose molecules forming long **unbranched chains** which are parallel to each other with cross linkages between them which give it its stability and a good structural material.

Structure:

- ❖ It is a polymer with straight chains of β -glucose units held by **glycosidic bonds** with the **OH group** projecting **out wards** from each chain forming cross linkages of hydrogen bonds with adjacent chains.
- ❖ The cross linking binds the chains together which associate to form **micro fibrils** that are arranged in larger bundles to form macro fibrils.

Structure of cellulose



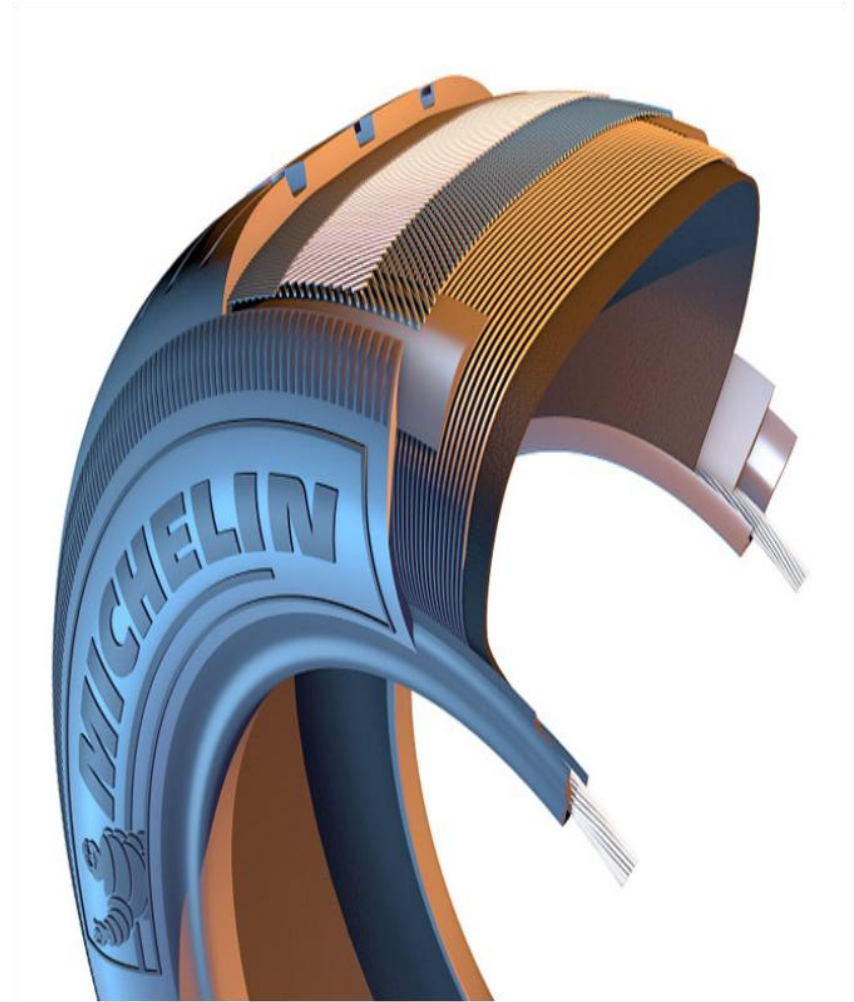
Take note

- ❖ starch lacks some structural properties possessed by cellulose because it lacks **cross linkages**.
- ❖ The stability of cellulose makes it difficult to digest and therefore not a good source of food to animals except those which have **cellulase** producing microorganisms which live in them in a symbiotic way e.g. in the rumen of cattle, goats, sheep, etc.

Uses of cellulose

1. Rayon

produced from cellulose extracted from wood is used in the manufacture of **tyre cords**.



Uses of cellulose

2. Cotton is used in the manufacture of **fabrics** and **clothes**.



Uses of cellulose

3. Cellophane
used in
packaging is
produced
from
cellulose.



Uses of cellulose

4. Paper is a product of cellulose.



Uses of cellulose

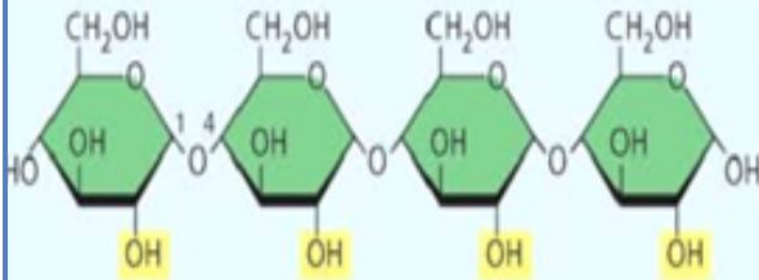
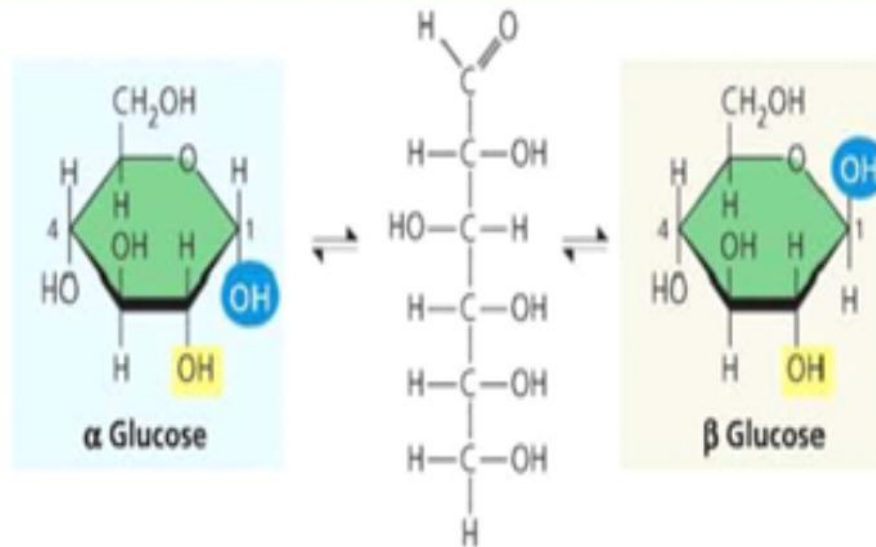
5. Celluloid
used in
***photographic
films*** is also a
derivative of
cellulose



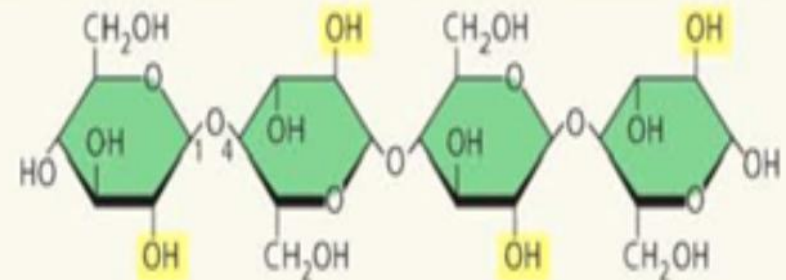
Polysaccharide structures

α and β glucose ring structures.

These two interconvertible forms of glucose differ in the placement of the hydroxyl group (highlighted in blue) attached to the number 1 carbon.



Starch: 1-4 linkage of α glucose monomers. All monomers are in the same orientation. Compare the positions of the —OH groups highlighted in yellow with those in cellulose (c).



(c) Cellulose: 1-4 linkage of β glucose monomers. In cellulose, every other β glucose monomer is upside down with respect to its neighbors.

Enzymes that digest starch by hydrolyzing its α linkages are unable to hydrolyze the β linkages of cellulose because of the distinctly different shapes of these two molecules. In fact, few organisms possess enzymes that can digest cellulose. Humans do not; the cellulose in our food passes through the digestive tract and is eliminated with the feces. Along the way, the cellulose abrades the wall of the digestive tract and stimulates the lining to secrete mucus, which aids in the smooth passage of food through the tract. Thus, although cellulose is not a nutrient for humans, it is an important part of a healthful diet. Most fresh fruits, vegetables, and whole grains are rich in cellulose. On food packages, “insoluble fiber” refers mainly to cellulose.

Some prokaryotes can digest cellulose, breaking it down into glucose monomers. A cow harbors cellulose-digesting prokaryotes in its rumen, the first compartment in its stomach (**Figure 5.9**). The prokaryotes hydrolyze the cellulose of hay and grass and convert the glucose to other nutrients that nourish the cow. Similarly, a termite, which is unable to digest cellulose by itself, has prokaryotes living in its gut that can make a meal of wood. Some fungi can also digest cellulose, thereby helping recycle chemical elements within Earth's ecosystems.

QUESTION

***HOW DOES THE STRUCTURE OF
CELLULOSE RELATE TO ITS
ROLES?***

SOLUTION TO THE QUESTION

*Cellulose is a polymer of straight chains of β -glucose units held by glycosidic bonds with OH groups projecting outwards from each chain forming **cross linkages** of H bonds with adjacent chains.*

- ❖ The **cross linking** binds the chains together offering much **tensile strength**.
- ❖ The **micro fibrils** in cell walls are arranged in several layers offering protection to the plant cell preventing it from bursting when water enters by osmosis.

SOLUTION TO THE QUESTION

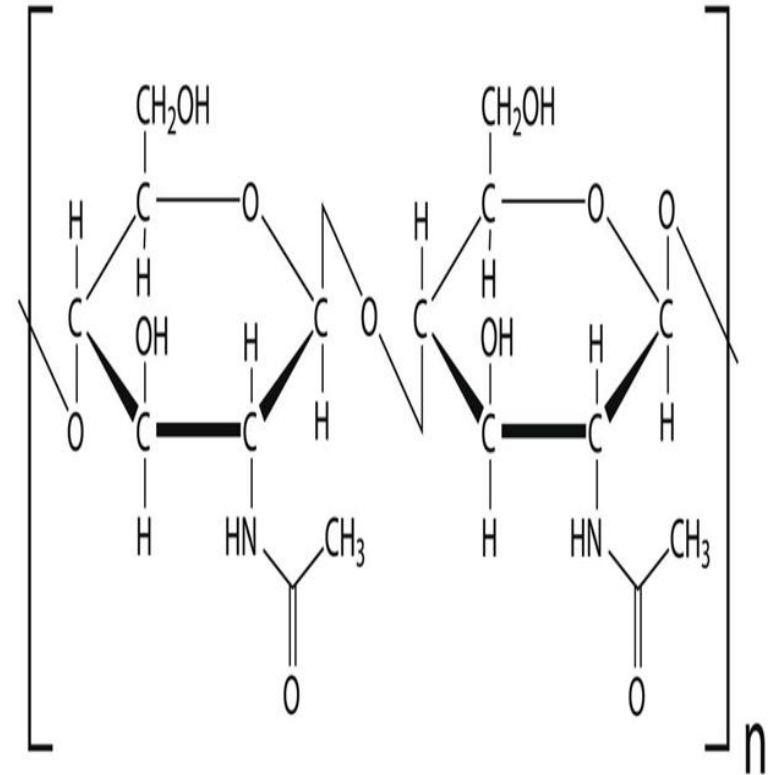
- ❖ The arrangement of **microfibrils** in cell walls contributes to **turgidity** hence offering support.
- ❖ The **parallel layers** of cellulose are fully permeable to water and **solutes**.
- ❖ The arrangement of **microfibrils** determines the shape of the cells and hence plant organs since it determines the direction in which cells expand as they grow.
- ❖ The **glycosidic bonds** holding the β -glucose units in cellulose can be **broken down** in presence of **enzyme** cellulase so that a free glucose molecule can be respired.

OTHER POLYSACCHARIDES:

Chitin

Chemically and structurally, chitin resembles cellulose but differs in possessing an **acetyl group** (NH-OCH₃) instead of one of the OH groups.

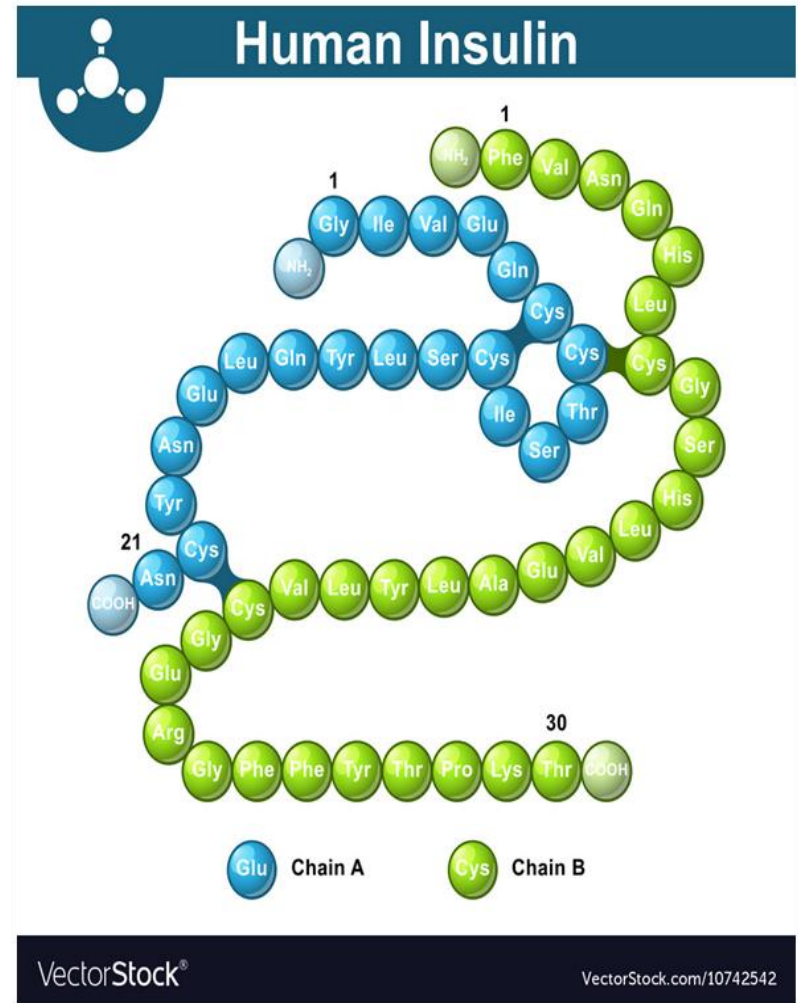
Like cellulose, it has a structural function and is a major component of **exoskeleton** of insects and crustacea. It is also found in fungal cell walls.



OTHER POLYSACCHARIDES

Insulin:

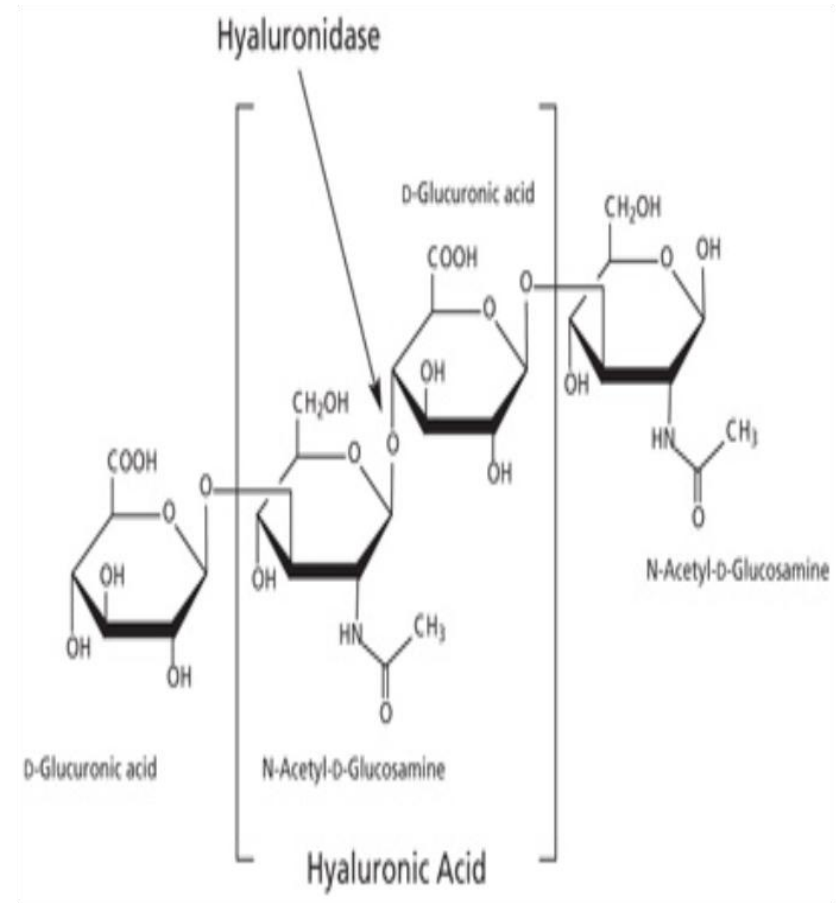
It is a *polymer* of **fructose** and found as a storage carbohydrate in some plants.



OTHER POLYSACCHARIDES

Mucopolysaccharides:

This group includes **hyaruronic acid** which forms part of the matrix of vertebrae connecting tissue. It is found in **cartilage, bones, vitreaous humor** of the eye, **synovial fluid**, etc.



OTHER POLYSACCHARIDES

Heparin:

An anti-coagulant also contains mucopolysaccharides

